Assignment 2

Problem 1

Generate a 3×3 matrix $\boldsymbol{A}$ of random integers that are between 7 and 17 (included).

a. Calculate the determinant of $\boldsymbol{A}$.
b. Calculate the rank of $\boldsymbol{A}$.
c. Show $\boldsymbol{A}$ in reduced-row echelon form. (Look for the command **rref**)
d. Calculate the eigenvectors and eigenvalues of $\boldsymbol{A}$.
e. Get the norm of each eigenvector.
f. Perform a sanity check by testing if $\boldsymbol{A} * \boldsymbol{V} - \boldsymbol{V} * \boldsymbol{D} = \boldsymbol{0}$, where * stands for matrix multiplication.
g. Calculate the E-B-E exponential and matrix exponential of $\boldsymbol{A}$.

Problem 2

Solve the following second-order differential equation:
$\ddot{y} - 6\dot{y} + 2y = e^t$, subject to the initial conditions $y(0) = 4$ and $\dot{y}(0) = 5$.

Problem 3

a. Calculate the indefinite integral of $\frac{1}{1+t^2}$

b. Calculate the following integral: $\int_{-1}^{1} e^{\frac{-x^2}{2}}\, dx$. Convert this result to double format.

Problem 4

Look for the command **symsum** and use it to calculate the following summations:

a. $x = \sum_{n=1}^{\infty} \frac{4(-1)^{n-1}}{2n-1}$

b. $y = \sum_{n=0}^{\infty} \frac{1}{\Gamma(n+1)}$ Use the command **gamma(n+1)**

Convert the results to double format.

Problem 5

Calculate the Laplace Transform of the following functions:

a. $f(t) = \lambda e^{-\lambda t}$

b. $g(t) = \cos(at)\, e^{-bt}$

c. Evaluate the values of $sF(s)$ and $sG(s)$ where $s = 0$ and $s = \infty$. (Use the commands **subs** and **limit**)

Problem 6

Using mesh analysis, get the values of I_1 and I_2. Show how you prepared the matrices in your report.

